Dynamic Programming: Floyd Warshall Algorithm

Session No.:
Course Name: Design and analysis of algorithm
Course Code: R1UC407B
Instructor Name: Mili Dhar
Duration: 50 Min.
Date of Conduction of Class:

Review of the key concepts

1. Review of previous session

Q: In a city map where each intersection is a node and roads are weighted edges, how would you find the shortest travel distance between every pair of intersections?

Learning Outcome

Apply the algorithm to solve all pair shortest path problems

Analyse the time complexity of the algorithm.

Session Outline

1 Introduction to Floyd Warshall Algorithm

2 Step by Step demonstration of Floyd warshall approach

3 Apply the method to find all pair shortest path

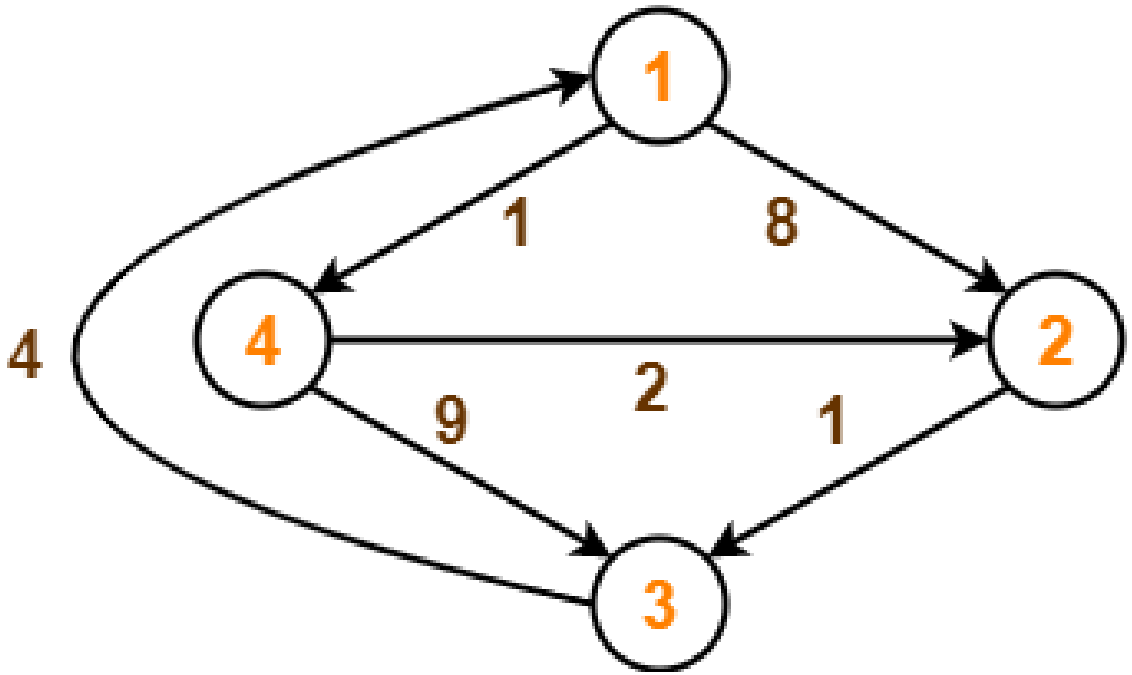
4 Analyse its complexity

Floyd Warshall (All pair Shortest Path)

1. An Algorithm for finding shortest paths in a weighted graph with positive or negative weights.
2. No negative cycle
3. Find the lengths of the shortest paths between all pairs of vertices

Floyd Warshall (Example)

Consider the following directed weighted graph Using Floyd Warshall Algorithm, find the shortest path distance between every pair of vertices.



Floyd Warshall (Solution)

Step 1-

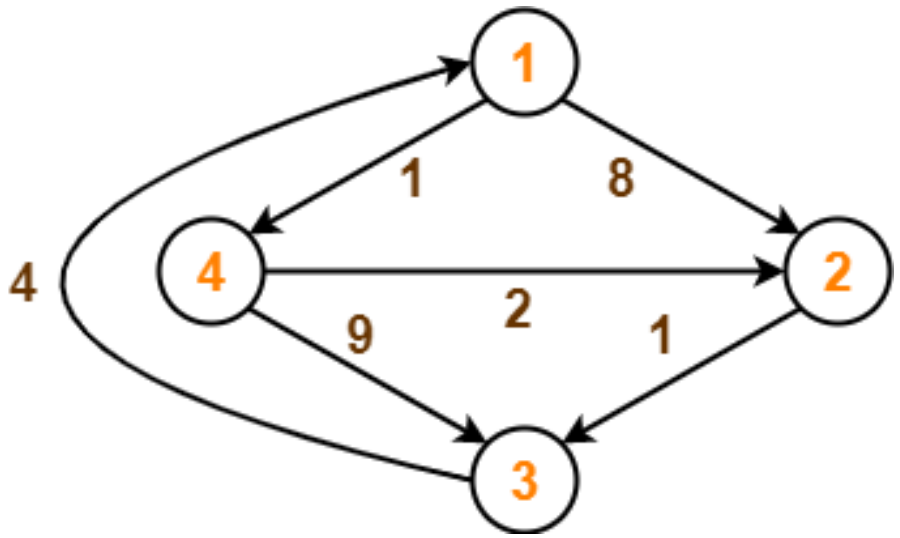
- Remove all the self loops and parallel edges (keeping the lowest weight edge) from the graph.
- In the given graph, there are neither self edges nor parallel edges.

Step 2-

- Write the initial distance matrix.
- It represents the distance between every pair of vertices in the form of given weights.
- For diagonal elements (representing self-loops), distance value = 0.
- For vertices having a direct edge between them, distance value = weight of that edge.
- For vertices having no direct edge between them, distance value = ∞.

Floyd Warshall (Solution)

Initial distance matrix for the given graph is-



$$D_0 = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 8 & \infty & 1 \\ \infty & 0 & 1 & \infty \\ 4 & \infty & 0 & \infty \\ \infty & 2 & 9 & 0 \end{bmatrix} \end{array}$$

Floyd Warshall (Solution)

Step-03: Using Floyd Warshall Algorithm, write the following 4 matrices. Here, we will consider intermediate nodes as k=1,2,3,4. When, k=1, we will keep 1st row and column as it is and k=2 means, 2nd row and column will be as it is and so on. For, rest of the positions, we need to consider the following formula.

if ($D_k[i,j] > D_{(k-1)}[i,k] + D_{(k-1)}[k,j]$)
$D_k[i,j] = D_{(k-1)}[i,k] + D_{(k-1)}[k,j]$
else
$D_k[i,j] = D_{(k-1)}[i,j]$

$$D_1 = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 8 & \infty & 1 \\ 2 & \infty & 0 & 1 & \infty \\ 3 & 4 & 12 & 0 & 5 \\ 4 & \infty & 2 & 9 & 0 \end{array}$$

$$D_2 = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 8 & 9 & 1 \\ 2 & & & & \\ 3 & 4 & 12 & 0 & 5 \\ 4 & \infty & 2 & 3 & 0 \end{array}$$

Floyd Warshall Algorithm | Problem-0

$$D_3 = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 8 & 9 & 1 \\ 2 & 5 & 0 & 1 & 6 \\ 3 & 4 & 12 & 0 & 5 \\ 4 & 7 & 2 & 3 & 0 \end{array}$$

$$D_4 = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 3 & 4 & 1 \\ 2 & 5 & 0 & 1 & 6 \\ 3 & 4 & 7 & 0 & 5 \\ 4 & 7 & 2 & 3 & 0 \end{array}$$

Floyd Warshall (Solution)

The last matrix D4 represents the shortest path distance between every pair of vertices.

Time complexity of Floyd Warshall algorithm is

$$\Theta(V \times V^2) = \Theta(V^3)$$

Thank You